Mureșan Raul-Călin

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Personal Website
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EXPERIENCE

Data Analyst (Working Student) | Robert Bosch *Oct 2025 – Present*

- **Engineered** an automated reminder system replacing manual follow-ups, reducing administrative workload for training compliance by **96%**.
- **Developed** a scalable data model using advanced DAX measures to map hierarchical structures across 3 management tiers.
- **Built** a real-time dashboard for training tracking, saving **~180 hours annually** by automating manual reporting tasks.

Programming Tutor | IT Junior *Jun 2024 – Sep 2025 (1y 4m)*

- **Mentored** 22 students in C, C++, HTML, CSS and Python, adapting complex curriculum to individual learning speeds and skill levels.
- **Designed** practical projects (Basic AI implementations, games, websites) to increase student engagement and technical retention.

EDUCATION

Technical University of Cluj-Napoca | B.Sc. in Computer Science *2022 – 2026 (Expected Graduation)*

- **Related Coursework:** Data Structures & Algorithms, Machine Learning, OOP, Artificial Intelligence.

PROJECTS

Automated Exam Digitization (Licence Thesis) | Python, OpenCV, Tesseract OCR

- **Engineered** a document digitization pipeline utilizing heuristic stitching algorithms and perspective correction to reconstruct fragmented questions.
- **Designed** a robust segmentation strategy (HSV masking, morphological dilation) to preserve complex chemical formulas where traditional OCR failed.
- **Integrated** Tesseract OCR for automated metadata extraction, reducing digitization workflow time by 90%.

NoteFlow | React, Next.js, Convex, TypeScript, Tailwind CSS

- **Developed** a content management app optimizing load times and SEO via Next.js **Server-Side Rendering (SSR)**.
- **Built** a type-safe, modular architecture using TypeScript to ensure scalability and reduce runtime errors.

Questify | React, PostgreSQL, REST API

- **Designed** a normalized PostgreSQL database schema to efficiently handle complex user interactions and voting logic.
- **Implemented** secure RESTful endpoints ensuring data integrity for a Q&A platform aimed at technical knowledge exchange.

IoT Security System | C++, Arduino UNO R4, IFTTT

- **Created** a motion-detection system integrating temperature/humidity sensors and real-time alerts.
- **Configured** IFTTT webhooks to trigger immediate email notifications based on specific sensor data thresholds.

Pacman Clone | Assembly x86

- **Engineered** a low-level game clone focusing on manual memory management and optimized graphics rendering logic.

SKILLS

Programming Languages: Python, C/C++, JavaScript/TypeScript, SQL, Assembly x86
Technologies: React, Next.js, PostgreSQL, Tailwind CSS, OpenCV, Arduino, Git, Linux
Languages: Romanian (Native), English (Fluent), French (Intermediate)

PROGRAMMING CONTESTS

Cloudflight Coding Contest (CCC) – 2023, 2024, 2025